MILITARY SPECIFICATION SHEET

POWDERS, PLASMA SPRAY, NICKEL ALUMINUM

POWDER, TYPE I, CLASS 2,

COMPOSITION G AND COMPOSITION H

This Interim Amendment is approved for use by the Naval Air Systems Command, Department of the Navy, with Military Specification Sheet MIL-P-83348/1, dated 25 October 1984.

PAGE 1

3.1, line 3 - delete "composition G powder" and substitute "compositions G and H powders".

3.1.1 - delete in its entirety and substitute "Metallographic Inspection." When inspected metallographically the powder particles for composition G material shall be capable of exhibiting a structure consisting of an aluminum spheroidal core essentially enclosed in a nickel shell and composition H shall be capable of exhibiting a spherical nickel core essentially surrounded by small aluminum particles".

3.1.2., 1st line - after "The" add "composition G."

PAGE 2

TABLE I - delete and substitute:

COATING PROPERTY	COATING COMPOSITION	Coating Process	
		THERMOSPRAY	PLASMASPRAY
COATING STRUCTURE	G	Dense sintered structure of nickel aluminides	
MELTING POINT	G	No appreciable melting below 3000°F	
BOND STRENGTH	G	3000 psi min.	5000 psi min.
	H		5000 psi min.
POROSITY	G	15% max.	15% max.
	H	15% max.	15% max.
UNMELTED PARTICLES	G		2% max.
	H		2% max.
HARDNESS	G		Rockwell 15T, 82min.
	H		Rockwell 15T, 82min.
DEPOSIT EFFICIENCY % minimum	G		N_2/H_2-55%, A/H_2-40%
	H		N_2/H_2-65%, A/H_2-60%

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Add: "3.4. Deposit efficiency. The deposit efficiency shall be as specified in Table I when determined by spraying a test panel not less than 2"X4" and determining the percent by weight of material deposited compared with that sprayed. The spray parameters and test equipment used in determining deposit efficiency shall be provided to the procuring activity by the contractor along with the test data".

6, 6.1, - Delete

Custodians:

Air Force - 20
Navy - AS
Army - MR

Review Activities:

Air Force - 99, 71
Navy - SH
Army - MR

Preparing activity:
Navy - AS
(Project No. MFFP-0334)

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